

Performance Testing

Date	29/09/2026
Team ID	UNM1425EduGenie
Project Name	EduGenie
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	EduGenie FastAPI learning activity APIs – /qa, /explain, /quiz, /summarize, /learn/recommendations
Test Environment	Local / Uvicorn Development Server
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Test Question Answering API with multiple student requests	5	60	API should return correct AI-generated responses without critical errors.
2	Test Topic Explanation API under repeated requests	5	60	Topic explanations should be generated successfully with acceptable response time.
3	Test Quiz Generation API under repeated requests	5	60	Quiz questions should be generated successfully without request failures.
4	Test Summarization and Learning Recommendation APIs	5	60	Summaries and recommendations should be generated successfully with stable performance.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	[Enter measured value]	[Pass / Fail]	Record the average response time from the testing tool.
2	Response Time (Max)	< 5 seconds	[Enter measured value]	[Pass / Fail]	Record the maximum response time observed.
3	Throughput (Req/sec)	[Enter target]	[Enter measured value]	[Pass / Fail]	Record the number of successfully processed requests per second.
4	Error Rate	< 1%	[Enter measured value]	[Pass / Fail]	Record failed requests as a percentage of total requests.
5	CPU Utilization	< 80%	[Enter measured value]	[Pass / Fail]	Record peak CPU utilization during testing.
6	Memory Utilization	< 80%	[Enter measured value]	[Pass / Fail]	Record peak memory utilization during testing.

Step 4: Observations & Analysis

The EduGenie application was tested under repeated requests to evaluate the performance of its learning activity APIs. The testing focuses on response time, throughput, error rate, CPU utilization, and memory utilization.

The Question Answering, Topic Explanation, Quiz Generation, Summarization, and Learning Recommendation features depend on the Google Gemini API. Therefore, response time may vary depending on external API response time and network conditions.

The final performance status should be determined using the actual measurements recorded during testing. Any performance issue identified during testing should be analyzed and optimized accordingly.

Step 5: Screenshots / Evidence

Attach screenshots of the Postman performance testing results, including:

1. API request and response results.
2. Response-time measurements.
3. Successful and failed requests.
4. Performance testing summary/results.
5. CPU and memory utilization, if measured.